

`usepackage[a4paper,margin=0.65in]geometry`

Arthur Cayley

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611. Report of the Committee on Mathematical Tables (continued)

[Continued from earlier pages of the same paper.]

[p. 479] where the second line 1, 13, 5, ... shows the numerical factors of the third column. The value of this second line as a result is not very obvious.

The third line gives x , and the fourth line y .

29. The second table, pp. 109–112, is entitled “Tabula II. Solutionem aequationis $y^2 - ax^2 + 1 = 0$, quotiescunque valor ipsius a talem admiserat, exhibens”; viz. it is remarked that this is only possible (but see *infra*) for those values of a which in Table I. correspond to a period of an even number of terms, as shown by two equal numbers in brackets; thus $a = 13$, the period of $\sqrt{13}$ given in Table I. is (1, 1, 1, 1) as shown by the top line 3, 1 (1, 1), and accordingly 13 is one of the numbers in Table II.; and we have there

$$13 \mid \frac{5}{18}.$$

Or take another specimen, $241 \mid \frac{4574225}{71011068}$, viz. the first line gives the value of x , and the second line the value of y (least values), for which $y^2 - ax^2 = -1$.

It is to be noticed that $a = 2$ and $a = 5$, for which we have obviously the solutions ($x = 1, y = 1$) and ($x = 1, y = 2$) respectively, are exceptional numbers not satisfying the test above referred to; and (apparently for this reason) the values in question, 2 and 5, are omitted from the table.

30. **Cayley**, “Table des plus petites solutions impaires de l’équation $x^2 - Dy^2 = \pm 4$, $D \equiv 5 \pmod{8}$.” *Crelle*, t. LIII. (1857), page 371 (one page), [231].

As regards the theory of quadratic forms, it is important to know whether for a given value of $D (\equiv 5, \pmod{8})$ there does or does not exist a solution, in odd numbers, of the equation $x^2 - Dy^2 = 4$. As remarked in the paper, “Note sur l’équation $x^2 - Dy^2 = \pm 4$, $D \equiv 5 \pmod{8}$,” pp. 369–371, [231], this can be determined for values of D of the form in question up to $D = 997$ by means of Degen’s Table; and the solutions, when they exist, of the equation $x^2 - Dy^2 = 4$, as also of the equation $x^2 - Dy^2 = -4$, can be obtained up to the same value of D . Observe that when the equation $x^2 - Dy^2 = -4$ is possible, the equation $x^2 - Dy^2 = +4$ is also possible, and that its least solution is obtained very readily from that of the other equation; it is therefore sufficient to tabulate the solution of $x^2 - Dy^2 = \pm 4$, the sign being $-$ when the corresponding equation is possible, and being in other cases $+$.

Hence the form of the Table: viz. as a specimen we have

D	$\pm$	x	y
757		imposs.	
765	+	83	3
773	–	139	5
781		imposs.	

[p. 480] that is, if $D = 757$ or 781 , there is no solution of either $x^2 - Dy^2 = +4$ or -4 ; if $D = 765$, there is a solution $x = 83$, $y = 3$ of $x^2 - Dy^2 = +4$, but none of $x^2 - Dy^2 = -4$; if $D = 773$, there is a solution $x = 139$, $y = 5$ of $x^2 - Dy^2 = -4$, and therefore also a solution of $x^2 - Dy^2 = +4$; and so in other cases.

[F. 14. *Partitions.*] Art. IV.

31. The problem of Partitions is closely connected with that of Derivations. Thus if it be asked in how many ways can the number n be expressed as a sum of three parts, the parts being 0, 1, 2, 3, and each part being repeatable an indefinite number of times, it is clear that n is at most = 9, and that for the values of n , = 0, 1, ..., 9 shown by the top line of the annexed table, the number of partitions has the values shown by the bottom line thereof:

0	1	2	3	4	5	6	7	8	9
a^3	a^2b	a^2c ab^2	a^2d abc b^3	abd b^2d b^2c	acd bcd bc^2	ad^2 c^2d c^3	bd^2 cd^2	cd^2	d^3
1	1	2	3	3	3	3	2	1	1

32. But taking a, b, c, d to stand for 0, 1, 2, 3 respectively, the actual partitions of the required form are exhibited by the literal terms of the table (these being obtained, each column from the preceding one, by the method of derivations, or say by the rule of the last and last but one), and the numbers of the bottom line are simply the number of terms in the several columns respectively.

A set of such literal tables, say of tables $\left(\begin{smallmatrix} a, b, c, \dots, k \\ 0, 1, 2, \dots, m \end{smallmatrix} \right)^n$ for different values of n and m (where the number of letters is = $m + 1$), would be extremely interesting and valuable. The tables for a given value of m and for different values of n are, it is clear, the proper foundation of the theory of the binary qantic $(a, b, c, \dots, k \not{x}, 1)^m$, which corresponds to such value of m . Prof. Cayley regrets that he has not in his covariant tables given in every case the complete series of literal terms; viz. the literal terms which have zero coefficients are, for the most part, though not always, omitted in the expressions of the several covariants.

33. But the question at present is as to the *number* of terms in a column, that is, as to the number of the partitions of a given form: the analytical theory has been investigated by Euler and others. The expression for the number of partitions is usually obtained as equal to the coefficient of x^n in the development, in ascending powers of x , of a given rational function of x : for instance, if there is no limitation as to the number of the parts, but if the parts are $1, 2, 3, \dots, m$ (viz. a part may have any value not exceeding m), each part being repeatable an indefinite number of times, then

Number of partitions of n = coefficient of x^n in $\frac{1}{(1-x)(1-x^2)(1-x^3)\dots(1-x^m)}$,

[p. 481] and we can, by actual development, obtain for any given values of m, n the number of partitions.

These have been tabulated $m = 1, 2, \dots, 20$, and $m = \infty$ (viz. there is in this case no limit as to the largest part), and $n = 1$ to 59, by

Euler, *Op. Arith. Coll.* t. I. pp. 97–101, given in the paper “De Partitione Numerorum,” pp. 73–101, (1750); the heading is “Tabula indicans quot variis modis numerus n e numeris $1, 2, 3, 4, \dots, m$, per additionem exhibi potest, seu exhibens valores formulae $n^{(m)}$.” The successive lines are, in fact, the coefficients of the several powers $x^0, x^1, \dots, x^n$ in the expansions of the functions

$$\frac{1}{1-x}, \quad \frac{1}{1-x \cdot 1-x^2}, \quad \dots, \quad \frac{1}{1-x \cdot 1-x^2 \dots 1-x^m}.$$

34. The generating function for any given value of m is, it is clear, $\frac{1}{1-x^m}$ multiplied by that for the next preceding value of m , and it thus appears how each line of the table is calculated from that which precedes it. The auxiliary numbers are printed; thus a specimen is

m	Valores numeri n .										
	0	1	2	3	4	5	6	7	8	9	10
4	1	1	2	3	5	6	9	11	15	18	23
					1	1	2	3	5	7	9
5	1	1	2	3	5	7	10	13	18	23	30

viz. suppose the numbers in the second 4-line known: then simply moving these each five steps onward we have the (auxiliary) numbers of the first 5-line; and thence by a mere addition the required series of numbers shown by the second 5-line. And similarly from this is obtained the second 6-line, and so on.

35. More extensive tables are contained in the memoir by

Marano, *Sulle leggi delle derivate generali delle funzioni di funzioni et sulla teoria delle forme di partizione dei numeri interi*, (4°. Genova, 1870),

pp. 1–281; and three tables paged separately, described merely as “Tavole dei numeri $C_{q,r}$, $S_{q,r}$, $S'_{q,r}$ citate nel testo colle indicazioni di Tavole I., II., III., ai n. 77, 79, 81”; viz. the reader is referred to these articles for the explanations of what the tabulated functions are; and there is not even then any explicit statement, but the investigation itself has to be studied to make out what the tables are. It is, in fact, easier to make this out from the tables themselves; the explanation is as follows:

[p. 482] Table I. (16 pages) is, in fact, Euler’s table, showing in how many ways the number n can be made up with the parts $1, 2, 3, \dots, m$; but the extent is greater, viz. n is from 1 to 103, and m from 1 to 102. The auxiliary numbers given in Euler’s table are omitted, as also certain numbers which occur in each successive line; thus a specimen is

$n =$	1	2	3	4	5	6	7	8	9	10 &c.
$C_{n,1}$	1	0	0	0	0	0	0	0	0	0
$C_{n,2}$		1	1	1	1	1	1	1	1	1
$C_{n,3}$			2	2	3	3	4	4	5	5
$C_{n,4}$				3	4	5	7	8	10	12
$C_{n,5}$					5	6	9	11	15	18

where the line $C_{n,4}$, (ways of making up $n - 1$ with the parts $1, 2, 3, 4$) is $1, 1, 2, 3, 5, 6, 9, 11, 15, 18$, &c., viz. we read from the corner diagonally downwards as far as the 6, and then horizontally along the line: this saves a large number of figures. The table is printed in ordinary quarto pages, which are taken to come in in tiers of seven, five, and three pages one under the other, as shown by a prefixed diagram; and the necessity of a large folding plate is thus avoided.

The successive lines give, in fact, the coefficients in the expansions of

$$\frac{1}{1-x}, \quad \frac{1}{1-x \cdot 1-x^2}, \quad \frac{1}{1-x \cdot 1-x^2 \cdot 1-x^3}, \quad \dots, \quad \frac{1}{1-x \cdot 1-x^2 \dots 1-x^m},$$

each expanded as far as x^{102} .

Table II. (6 pages). The successive lines give the coefficients in the expansions of

$$S, \quad \frac{S}{1-x}, \quad \frac{S}{1-x \cdot 1-x^2}, \quad \dots, \quad \frac{S}{1-x \cdot 1-x^2 \dots 1-x^m},$$

where

$$S = \frac{1}{(1-x)(1-x^2)(1-x^3) \dots \text{ad inf.}},$$

each expanded as far as x^{50} , and further continued as regards the first ten lines, that is, the expansions of

$$S, \quad \frac{S}{1-x}, \quad \frac{S}{1-x \cdot 1-x^2}, \quad \dots, \quad \frac{S}{1-x \cdot 1-x^2 \dots 1-x^9},$$

each as far as x^{100} .

Table III. (2 pages). The successive lines give the coefficients in the expansions of

$$S^2, \quad \frac{S^2}{1-x}, \quad \frac{S^2}{1-x \cdot 1-x^2}, \quad \cdots, \quad \frac{S^2}{1-x \cdot 1-x^2 \cdots 1-x^m},$$

each expanded as far as x^{60} .

[p. 483] 36. As regards Tables II. and III., the analytical explanations have been given in the first instance; but it is easy to see that the tables give numbers of partitions. Thus, in Table II., the second line gives the coefficients in the development of

$$\frac{1}{(1-x)^2(1-x^2)(1-x^3)\cdots},$$

viz. these are 1, 2, 4, 7, 12, 19, 30, ... being the number of ways in which the numbers 0, 1, 2, 3, 4, &c. respectively can be made up with the parts 1, 1', 2, 3, 4, &c.; thus

Partitions.	No. =
0	1
1 1'	2
2 1, 1 1, 1' 1', 1'	4
3 2, 1 2, 1' 1, 1, 1 1, 1, 1' 1, 1', 1' 1', 1', 1'	7
&c.	&c.

Similarly, the third line shows the number of ways in which these numbers respectively can be made up with the parts 1, 1', 2, 2', 3, 4, 5, &c.; the fourth line with the parts 1, 1', 2, 2', 3, 3', 4, 5, &c.; and so on.

And in like manner in Table III., the first line shows the number of ways when the parts are 1, 1', 2, 2', 3, 3', ...; the second line when they are 1, 1', 1'', 2, 2', 3, 3', &c.; the third when they are 1, 1', 1'', 2, 2', 2'', 3, 3', &c.; and so on.

It is clear that the series of tables might be continued indefinitely, viz. there might be a Table IV. giving the developments of

$$S^3, \quad \frac{S^3}{1-x}, \quad \frac{S^3}{1-x \cdot 1-x^2}, \quad \text{and so on.}$$

An interesting table would be one composed of the first lines of the above series, viz. a table giving in its successive lines the developments of S , S^2 , S^3 , S^4 , &c.

There are throughout the work a large number of numerical results given in a quasi-tabular form; but the collection of these, with independent explanations of the significations of the tabulated numbers, would be a task of considerable labour.

[p. 484]

[F. 15. *Quadratic forms $a^2 + b^2$, &c., and Partitions of Numbers into squares, cubes, and biquadrates.*] Art. V.

37. The forms here referred to present themselves in the various complex theories. Thus $N = a^2 + b^2 = (a + bi)(a - bi)$; this means that, in the theory of the complex numbers $a + bi$ (a and b integers), N is not a prime but a composite number.

It is well known that an ordinary prime number $\equiv 3, \text{ mod. } 4$, is not expressible as a sum $a^2 + b^2$, being, in fact, a prime in the complex theory as well as in the ordinary one: but that an ordinary prime number $\equiv 1, \text{ mod. } 4$, is (in one way only) $= a^2 + b^2$; so that it is in the complex theory a composite number. A number whose prime factors are each of them $\equiv 1, \text{ mod. } 4$, or which contains, if at all, an even number of times any prime factor $\equiv 3, \text{ mod. } 4$, can be expressed in a variety of ways in the form $a^2 + b^2$; but these are all easily deducible from the expressions in the form in question of its several factors $\equiv 1, \text{ mod. } 4$, so that the required table is a table of the form $p = a^2 + b^2$, p an ordinary prime number $\equiv 1 \text{ mod. } 4$: a and b are one of them odd, the other even; and to render the decomposition definite a is taken to be odd.

$p = a^2 + b^2$; viz. decomposition of the primes of the form $4n + 1$ into the sum of two squares: a table extending from $p = 5$ to 11981 (calculated by Zornow) is given by

Jacobi, *Crelle*, t. XXX. (1846), pp. 174–176.

This is carried by Reuschle, as presently mentioned, up to $p = 24917$. Reuschle notices that $2713 = 3^2 + 52^2$ is omitted, also $6997 = 39^2 + 74^2$, and that 8609 should be $= 47^2 + 80^2$.

38. Similarly, primes of the form $6n + 1$ are expressible in the form $p = a^2 + 3b^2$. Observe that, ω being an imaginary cube root of unity, this is connected with $p = (a + b\omega)(a + b\omega^2) = a^2 - ab + b^2$, viz. we have $4p = (2a - b)^2 + 3b^2$; or the form $a^2 + 3b^2$ is connected with the theory of the complex numbers composed of the cube roots of unity.

$p = a^2 + 3b^2$; viz. decomposition of the primes of the form $6n + 1$ into the form $p = a^2 + 3b^2$: a table extending from $p = 7$ to 12007 (calculated also by Zornow) is given by

Jacobi, *Crelle*, t. XXX. (1846), ut supra, pp. 177–179.

This is carried by Reuschle up to $p = 13369$, and for certain higher numbers up to 49999, as presently mentioned. Reuschle observes that $6427 = 80^2 + 3 \cdot 3^2$ is by accident omitted, and that 6481 should be $= 41^2 + 3 \cdot 40^2$.

39. Again, primes of the form $8n + 1$ are expressible in the form $p = a^2 + 2b^2$ (or say $= c^2 + 2d^2$), the theory being connected with that of the complex numbers composed with the 8th roots of unity (fourth root of -1 , $= \sqrt[4]{-1}$).

$p = c^2 + 2d^2$; viz. decomposition of primes of the form $8n + 1$ into the form $c^2 + 2d^2$:

[p. 485] a table extending from $p = 17$ to 5943 (extracted from a MS. table calculated by Struve) is given by

Jacobi, *Crelle*, t. XXX. (1846), ut supra, p. 180.

This is carried by Reuschle up to $p = 12377$, and for certain higher numbers up to 24889, as presently mentioned.

40. Reuschle's tables of the forms in question are contained in the work:

Reuschle, *Mathematische Abhandlung*, &c. (see ante No. 15), under the heading "B. Tafeln zur Zerlegung der Primzahlen in Quadrate" (pp. 22–41). They are as follows:

Table III. for the primes $6n + 1$.

The first part gives $p = A^2 + 3B^2$ and $4p = L^2 + 27M^2$, from $p = 7$ to 5743. The table gives A, B, L, M ; those numbers which have 10 for a cubic residue are distinguished by an asterisk. A specimen is

p	A	B	L	M
37*	5	2	11	1

viz. $37 = 5^2 + 3 \cdot 2^2$, $148 = 11^2 + 27 \cdot 1^2$; the asterisk shows that $x^3 \equiv 10 \pmod{37}$ is possible: in fact $34^3 \equiv 10 \pmod{37}$.

The second part gives $p = A^2 + 3B^2$ only, from $p = 5749$ to 13669. The table gives A, B ; and the asterisk implies the same property as before.

The third part gives $p = A^2 + 3B^2$, but only for those values of p which have 10 for a cubic residue, viz. for which $x^3 \equiv 10 \pmod{p}$ is possible, from $p = 13689$ to 49999. The table gives A, B ; the asterisk, as being unnecessary, is not inserted.

Table IV. for the primes $4n + 1$ in the form $A^2 + B^2$, and for those which are also $8n + 1$ in the form $C^2 + 2D^2$.

The first part gives $p = A^2 + B^2$, $= C^2 + 2D^2$, from $p = 5$ to 12377. The table gives A, B, C, D ; those numbers which have 10 for a biquadratic residue, viz. for which $x^4 \equiv 10 \pmod{p}$ is possible, are distinguished by an asterisk; those which have also 10 for an octic residue, viz. for which $x^8 \equiv 10$

(mod. p) is possible, by a double asterisk. A specimen is

p	A	B	C	D
229	15	2	— — —	— — —
233	13	8	15	2
241**	15	4	13	6

The second part gives $p = A^2 + B^2$, from $p = 12401$ to 24917 for all those values of p which have 10 for a biquadratic residue ($x^4 \equiv 10 \pmod{p}$ possible). The table gives A, B ; those values of p which have 10 for an octic residue, viz. for which $x^8 \equiv 10 \pmod{p}$ is possible, are distinguished by an asterisk.

[p. 486] The third part gives $p = C^2 + 2D^2$, from $p = 12641$ to 24889 for all those values of p which have 10 for a biquadratic residue. The table gives C, D ; those values of p which have 10 as an octic residue are distinguished by an asterisk.

41. A table by Zornow, *Crelle*, t. XIV. (1835), pp. 279, 280 (belonging to the Memoir “De Compositione numerorum e Cubis integris positivis,” pp. 276–280), shows for the numbers 1 to 3000 the least number of cubes into which each of these numbers can be decomposed. Waring gave, without demonstration, the theorem that every number can be expressed as the sum of at most 9 cubes. The present table seems to show that 23 is the only number for which the number of cubes is $= 9$ ($= 2 \cdot 2^3 + 7 \cdot 1^3$); that there are only fourteen numbers for which the number of cubes is $= 8$, the largest of these being 454; and hence that every number greater than 454 can be expressed as a sum of at most 7 cubes; and further, that every number greater than 2183 can be expressed as a sum of at most 6 cubes. A small subsidiary table (p. 276) shows that the number of numbers requiring 6 cubes gradually diminishes—e.g. between 12^3 and 13^3 there are seventy-five such numbers, but between 13^3 and 14^3 only sixty-four such numbers; and the author conjectures “that for numbers beyond a certain limit every number can be expressed as a sum of at most 5 cubes.”

42. For the decomposition of a number into biquadrates we have

Bretschneider, “Tafeln für die Zerlegung der Zahlen bis 4100 in Biquadrate,” *Crelle*, t. XLVI. (1853), pp. 3–23.

Table I. gives the decompositions, thus:

N	1^4	2^4	3^4	4^4	5^4
696	6	1	2	2	
$\vdots$					
3251			3	8	

viz. $696 = 6 \cdot 1^4 + 1 \cdot 2^4 + 2 \cdot 3^4 + 2 \cdot 4^4$, &c.

And Table II. enumerates the numbers which are sums of at least 2, 3, 4, ..., 19 biquadrates. There is at the end a summary showing for the

first 4100 numbers how many numbers there are of these several forms respectively: 28 numbers are each of them a sum of 2 biquadrates, 75 a sum of 3, ..., 7 a sum of 19 biquadrates. The seven numbers, each of them a sum of 19 biquadrates, are 79, 159, 239, 319, 399, 479, 559.

[F. 16. *Binary, Ternary, etc. quadratic and higher forms.*] Art. VI.

43. Euler worked with the quadratic forms $px^2 \pm qy^2$ (p and q integers), particularly in regard to the forms of the divisors of such numbers. It will be sufficient to refer to his memoir:

[p. 487] **Euler**, “Theoremata circa divisores numerorum in hac forma $pa^2 \pm qb^2$ contentorum,” (*Op. Arith. Coll.* pp. 35–61, 1744), containing fifty-nine theorems, exhibiting in a quasi-tabular form the linear forms of the divisors of such numbers. As a specimen:

“Theorema 13. Numerorum in hac forma $a^2 + 7b^2$ contentorum divisores primi omnes sunt vel 2, vel 7, vel in una sex formularum

$$\begin{array}{ll} 28m + 1, & 28m + 11, \\ 28m + 9, & 28m + 15, \\ 28m + 25, & 28m + 23, \end{array}$$

seu in una harum trium

$$14m + 1, \quad 14m + 9, \quad 14m + 11,$$

sunt contenti”; viz. the forms are the three $14m + 1$, $14m + 9$, $14m + 11$.

But Euler did not consider, or if at all very slightly, the trinomial forms $ax^2 + bxy + cy^2$, nor attempt the theory of the reduction of such forms. This was first done by Lagrange in the memoir

Lagrange, *Mém. de Berlin*, 1773. And the theory is reproduced by

Legendre, *Théorie des Nombres*, Paris, 1st ed. 1798; 2nd ed. 1808, § 8, “Réduction de la formule $Ly^2 + Myz + Nz^2$ à l’expression la plus simple,” (2nd ed. pp. 61–67).

44. But the classification of quadratic forms, as established by Legendre, is defective as not taking account of the distinction between proper and improper equivalence; and the ulterior theory as to orders and genera, and the composition of forms (although in the meantime established by Gauss), are not therein taken into account; for this reason the Legendre’s Tables I. to VIII. relating to quadratic forms, given after p. 480 (thirty-two pages not numbered), are of comparatively little value, and it is not necessary to refer to them in detail.

The complete theory was established by

Gauss, *Disquisitiones Arithmeticae*, 1801.

It is convenient to refer also to the following memoir

Lejeune Dirichlet, “Recherches sur diverses applications de l’Analyse à la théorie des Nombres,” *Crelle*, t. XIX. (1839), p. 338, [*Ges. Werke*, t. I. p. 427], as giving a succinct statement of the principle of classification, and in particular a table of the characters of the genera of the properly primitive order, according to the four forms $D = PS^2$, $P \equiv 1$ or $3 \pmod{4}$, and $D = 2PS^2$, $P \equiv 1$ or $3 \pmod{4}$, of the determinant.

45. Tables of quadratic forms arranged on the Gaussian principle are given by

Cayley, *Crelle*, t. LX. (1862), pp. 357–372, [335]; viz. the tables are —

Table I. des formes quadratiques binaires ayant pour déterminants les nombres négatifs depuis $D = -1$ jusqu’à $D = -100$. (Pp. 360–363: [*Coll. Math. Papers*, t. V, pp. 144–147].)

[p. 488] A specimen is

D	Classes	α	β	δ	ϵ	$\delta\epsilon$	Cp
$= 26$	1, 0, 26	+	+			+	1
	3, -1, 9	+	*			+	g^2
	3, 1, 9	+				+	g^4
	5, 2, 6						g
	2, 0, 13	—				—	g^3
	5, -2, 6	—				—	g^5

where α, β denote, as there explained, the characters in regard to the odd prime factors of D ; $\delta, \epsilon, \delta\epsilon$ those in regard to the numbers 4 and 8. The last column shows that the forms in the two genera respectively are $1, g^2, g^4$ and g, g^3, g^5 , where $g^6 = 1$, viz. the form g , six times compounded, gives the principal form $(1, 0, 26)$.

Table II. des formes quadratiques binaires ayant pour déterminants les nombres positifs non-carrés depuis $D = 2$ jusqu’à $D = 99$. (Pp. 364–369: [*l.c.*, pp. 148–153].)

The arrangement is the same, except that there is a column “Périodes” showing, in an easily understood abbreviated form, the period of each form. Thus $D = 7$, the period of the principal form $(1, 0, -7)$, is given as $1, 7, -3, \ddot{1}, 2, \ddot{1}, -3, 7, 1$, which represents the series of forms $(1, 2, -3)$, $(-3, 1, 2)$, $(2, 1, -3)$, $(-3, 2, 1)$.

Table III. des formes quadratiques binaires pour les treize déterminants négatifs irréguliers du premier millier. (Pp. 370–372: [*l.c.*, pp. 154–156].)

The arrangement is the same as in Table I. It may be mentioned that the thirteen numbers, and the forms for the principal genus for these numbers, respectively are:

	Principal genus
576, 580, 820, 900	$(1, e^2)(1, e_1^2)$
884	$(1, e^2)(1, i^2, i^4, i^6)$
243, 307, 339, 459, 675, 891	$(1, d, d^2)(1, d_1, d_1^2)$
755, 974	$(1, d, d^2)(1, d_1, d_1^2)(1, e^2)$

where $d^3 = d_1^3 = 1$, $e^4 = e_1^4 = 1$, $i^8 = 1$, viz. $(1, e^2)(1, e_1^2)$ denotes four forms, $1, e^2, e_1^2, e^2 e_1^2$; and so in the other cases.

46. Gauss must have computed quadratic forms to an enormous extent; but, for the reasons (rather amusing ones) mentioned in a letter of May 17, 1841, to Schumacher (quoted in Prof. Smith's Report on "The Theory of Numbers," Brit. Assoc. Report for 1862, p. 526, [and Smith's *Coll. Math. Papers*, t. I. p. 261]), he did not preserve his results in detail, but only in the form appearing in the

"Tafel der Anzahl der Classen binärer quadratischer Formen," *Werke*, t. II. pp. 449–476; see editor's remarks, pp. 521–523.

[p. 489] This relates almost entirely to negative determinants, only three quarters of p. 475 and p. 476 to positive ones; for negative determinants, it gives the number of genera and classes, as also the index of irregularity for the determinants of the hundreds 1 to 30, 43, 51, 61, 62, 63, 91 to 100, 117 to 120; then, in a different arrangement, for the thousands 1, 3 and 10, for the first 800 numbers of the forms $-(15n+7)$ and $-(15n+13)$; also for some very large numbers, and for positive determinants of the hundreds 1, 2, 3, 9, 10, and for some others.

A specimen is

Centas I.	
G II. (58) ... (280)	
1.	5, 6, 8,
	9, 10, 12,
	13, 15, 16,
	18, 22, 25,
	28, 37, 58,
2.	14, 17, 20,
Summa 233 ... 477	
Irreg.	0 Impr. 74

viz. this shows, as regards the negative determinants 1 to 100, that the determinants belonging to G II. 1, viz. those which have two genera each of one class, are 5, 6, 8, 9, &c., in all fifteen determinants; those belonging to G II. 2, viz. those which have two genera each of two classes, are 14, 17, 20, &c.; and so on. The head numbers (58) ... (280) show the number of determinants,

each having two genera, and the number of classes; thus,

$$\begin{array}{rcl}
 \text{G II.} & 1 \times 15 = & 15 \\
 & 2 \times 17 = & 34 \\
 & 3 \times 17 = & 51 \\
 & 4 \times 6 = & 24 \\
 & 5 \times 2 = & 10 \\
 & 6 \times 1 = & 6 \\
 \hline
 & 58 & 140 \\
 & & \times 2 \\
 & & \hline
 & & = 280;
 \end{array}$$

and the bottom numbers show the total number of genera and of classes, thus

$$\begin{array}{rcl}
 \text{G I.} & 17 \times 1 = 17 & 61 \\
 \text{II.} & 58 \times 2 = 116 & 280 \\
 \text{IV.} & 25 \times 4 = 100 & 136 \\
 \hline
 & 100 & 233 & 477
 \end{array}$$

[p. 490] viz. seventeen determinants, each of one genus, and together of sixty-one classes; fifty-eight determinants, each of two genera, and together of 280 classes; and twenty-five determinants, each of four genera, and together 136 classes, give in all 233 genera and 477 classes. These are exclusive of 74 classes belonging to the improperly primitive order; and the number of irregular determinants (in the first hundred) is = 0.

The irregular determinants are indicated thus:

243(*3*),
 307(*3*), 339(*3*),
 459(*),
 576(*2*), 580(*2*),
 675(*3*),
 755(*3*),
 891(*3*), 820(*2*), 900(*2*), 884(*2*), 974(*3*),
 *3 * 243, 307, 339, 459?, 675, 755, 891,
 *2 * 576, 580, 820, 884, 900, 974,

which is a notation not easily understood.

As regards the positive determinants, a specimen is

Centas I.
Excedunt determinantis
quadrati 10.
G I. ... (12),
1. 2, 5, 13,
17, 29, 41,
53, 61, 73,
89, 97,
3. 37

viz. in the first hundred, the positive determinants having one genus of one class are 2, 5, 13, &c. ... (eleven in number); that having one genus of three classes is 37, (one in number); $11 + 1 = 12$. The irregular determinants, if any, are not distinguished.

47. *Binary cubic forms*. — The earliest table is given by

Arndt, “Tabelle der reducirten binären kubischen Formen und Klassen für alle negativen Determinanten $-D$ von $D = 3$ bis $D = 2000$,” *Grunert’s Archiv*, t. XXXI. 1858, pp. 369–448.

The memoir is a sequel to one in t. XVII. (1851). The binary cubic form (a, b, c, d) , of determinant $D (= (bc - ad)^2 - 4(b^2 - ac)(c^2 - bd))$, is said to be reduced when its characteristic $\phi, = (A, B, C), = (2(b^2 - ac), bc - ad, 2(c^2 - bd))$, is a reduced quadratic form, that is, when in regard to absolute values B is not $> \frac{1}{2}A$, C not $< A$.

[p. 491] A specimen is

D	Reduced forms, with characters	Classes
44	$(0, 1, 0, -11)$ $(1, -1, -2, 0)$ $(2, 0, 22)$	$(0, -1, 0, 11)$ $(0, -2, 1, 1)$ (6, 8)

Two subsidiary tables are given, pp. 351, 352, and 353–368.

48. It appeared suitable to remodel a part of this table in the manner made use of for quadratic forms in my tables above referred to; and it is accordingly divided into the three tables given by

Cayley, *Quart. Math. Journ.* t. XI. (1871), where the notation &c. is explained, pp. 251–261. [496]; viz. these are: —

Table I. of the binary cubic forms, the determinants of which are the negative numbers $\equiv 0 \pmod{4}$ from -4 to -400 (pp. 251–258 [*Coll. Math. Papers*, t. VIII., pp. 55–61]).

A specimen is

Det. 4×-11 .	Classes.	Order.	Charact.	Comp.
	0, 1, 0, -11	on	1, 0, 11	1
	0, 2, -1, -2	pp pp	3, 1, 4	d
	0, 2, 1, 1		3, -1, 4	d^2

Table II. of the binary cubic forms the determinants of which (taken positively) are $\equiv 1 \pmod{4}$ from -3 to 99 , the original heading is here corrected, [*l.c.*, pp. 61, 62]; and

Table III. of the binary cubic forms the determinants of which are the negative numbers -972 , -1228 , -1336 , -1836 , and -2700 , [*l.c.*, pp. 63, 64]; viz. $-972 = 4 \times -243$, $\dots$, $-2700 = 4 \times -675$, where -243 , $\dots$, -675 are the first six irregular numbers for quadric forms.

4×-675 , $= -2700$ is beyond the limits of Arndt's tables, and for this number the calculation had to be made anew; the table gives nine classes $(1, d, d^2)(1, d_1, d_1^2)$ of the order *ip* on *pp*, but it is remarked that there may possibly be other cubic classes based on a non-primitive characteristic; the point was left unascertained.

49. The theory of ternary quadratic forms was discussed and partially established by Gauss in the *Disquisitiones Arithmeticae*. It is proper to recall that a ternary quadratic form is either determinate, viz. always positive, such as $x^2 + y^2 + z^2$, or always negative, such as $-x^2 - y^2 - z^2$; or else it is indeterminate, such as $x^2 + y^2 - z^2$. But as regards determinate forms, the negative ones are derived from the positive ones by simply reversing the signs of all the coefficients, so that it is sufficient to attend to the positive forms; and practically the two cases are positive forms (meaning thereby positive determinate forms) and indeterminate forms; but the theory for positive forms was first established completely, and so as to enable the formation of tables, in the work

Seeber, *Ueber die Eigenschaften der positiven ternären quadratischen Formen*, (4to. Freiburg, 1831),

[p. 492] which is reviewed by Gauss in the *Gött. Gelehrte Anzeigen*, 1831, July 9 (see Gauss, *Werke*, t. II. pp. 188–193). The author gives (pp. 220–243) tables “of the classes of positive ternary forms represented by means of the corresponding reduced forms” for the determinants 1 to 100. A specimen is

$$\text{Det. 6} \quad \begin{pmatrix} 1, 1, 6 \\ 0, 0, -1 \end{pmatrix}, \begin{pmatrix} 1, 2, 3 \\ -1, -1, 0 \end{pmatrix}.$$

$$\text{Zugeordnete Formen} \quad \begin{pmatrix} 8, 8, 3 \\ 0, 0, 4 \end{pmatrix}, \begin{pmatrix} 7, 7, 4 \\ 4, 4, 2 \end{pmatrix}.$$

where it is to be observed that Seeber admits odd coefficients for the terms in yz , zx , xy ; viz. his symbol $\begin{pmatrix} a, b, c \\ f, g, h \end{pmatrix}$ denotes

$$ax^2 + by^2 + cz^2 + fyz + gzx + hxy,$$

and his determinant is

$$4abc - af^2 - bg^2 - ch^2 + fgh.$$

Also his adjoint form is

$$\begin{pmatrix} 4bc - f^2, & 4ca - g^2, & 4ab - h^2 \\ 2gh - 4af, & 2hf - 4bg, & 2fg - 4ch \end{pmatrix}, \quad = (4bc - f^2)x^2 + \dots + (2gh - 4af)yz + \dots$$

In the notation of the *Disquisitiones Arithmeticae*, followed by Eisenstein and others, the symbol $\begin{pmatrix} a, b, c \\ f, g, h \end{pmatrix}$ denotes

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy;$$

the determinant is

$$= -(abc - af^2 - bg^2 - ch^2 + 2fgh),$$

a positive form having thus always a negative determinant. And the adjoint form is

$$-\begin{pmatrix} bc - f^2, & ca - g^2, & ab - h^2 \\ gh - af, & hf - bg, & fg - ch \end{pmatrix} = -(bc - f^2)x^2 - \dots - 2(gh - af)yz - \dots$$

Hence Seeber's determinant is $= -4$ multiplied by that of Gauss, and his tables really extend between the values -1 and -25 of the Gaussian determinant.

50. Tables of greater extent, and in the better form just referred to, are given by

Eisenstein, *Crelle*, t. XLI. (1851), pp. 169–190; viz. these are

I. “Tabelle der eigentlich primitiven positiven ternären Formen für alle negativen Determinanten von -1 bis -100 ,” (pp. 169–185).

A specimen is

D	Anzahl	Reducirte Formen für $-D$
10	3	$\begin{pmatrix} 1, 1, 10 \\ 0, 0, 0 \end{pmatrix}, \begin{pmatrix} 1, 2, 5 \\ 0, 0, 0 \end{pmatrix}, \begin{pmatrix} 2, 2, 3 \\ 0, -1, 0 \end{pmatrix}$ $\delta = 8 \quad \delta = 4 \quad \delta = 4$

[p. 493] II. “Tabelle der uneigentlich primitiven positiven ternären Formen für alle negativen Determinanten von -2 bis -100 ,” (pp. 186–189).

A specimen is

D	Anzahl	Reducirte Formen für $-D$
10	1	$\begin{pmatrix} 1, 2, 4 \\ 1, 1, 1 \end{pmatrix}$ $\delta = 6$

And there is given (p. 190) a table of the reduced forms for the determinant $-385 (= -5 \cdot 7 \cdot 11)$, selected merely as a largish number with three factors; viz. there are in all fifty-nine forms, corresponding to values 1, 2, 4, 6, 8 of δ .

It may be remarked that δ denotes, for any given form, the number of ways in which this is linearly transformable into itself, this number being always 1, 2, 4, 6, 8, 12, or 24. The theory as to this and other points is explained in the memoir (pp. 141–168), and various subsidiary tables are contained therein and in the *Anhang* (pp. 227–242); and there is given a small table relating to indeterminate forms, viz. this is

“C. Versuch einer Tabelle der nicht äquivalenten unbestimmten (indifferenten) ternären quadratischen Formen für die Determinanten ohne quadratischen Theiler unter 20,” (pp. 239, 240).

A specimen is

D	Indifferente ternäre quadratische Formen
10	$\begin{pmatrix} 0, 1, 10 \\ 0, 0, 1 \end{pmatrix}, \begin{pmatrix} 1, 2, 5 \\ 0, 0, 0 \end{pmatrix}$ $\begin{pmatrix} 0, 0, 10 \\ 0, 0, 1 \end{pmatrix}$

where, when the determinant is even, the forms in the second line are always improperly primitive forms.

[F. 17. *Complex Theories.*] Art. VII.

51. The theory of binary quadratic forms (a, b, c) , with complex coefficients of the form $\alpha + \beta i$, ($i = \sqrt{-1}$ as usual, α and β integers), has been studied by Lejeune Dirichlet, Prof. H. J. S. Smith, and possibly others; but no tables have, it is believed, been calculated. The calculations would be laborious; but tables of a small extent only would be a sufficient illustration of the theory, and would, it is thought, be of great interest.

[p. 494] The theory of complex numbers of the last-mentioned form $\alpha + \beta i$, or say of the numbers formed with the fourth root of unity, had previously been studied by Gauss; and the theory of the numbers formed with the cube roots of unity ($\alpha + \beta \omega$, $\omega^3 + 1 = 0$, α and β integers) was studied by Eisenstein; but the general theory of the numbers involving the n th roots of unity (λ an odd prime) was first studied by Kummer. It will be sufficient to refer to his memoir,

Kummer, “Zur Theorie der complexen Zahlen,” *Berl. Monatsb.*, March, 1845; and *Crelle*, t. XXXV. (1847), pp. 319–326; also “Ueber die Zerlegung der aus Wurzeln der Einheit gebildeten complexen Zahlen in ihre Primfactoren,” same volume, pp. 327–367, where the astonishing theory of “Ideal Complex Numbers” is established.

52. It may be recalled that, p being an odd prime, and ρ denoting a root of the equation $\rho^{p-1} + \rho^{p-2} + \dots + \rho + 1 = 0$, then the numbers in question are those of the form $a + b\rho + \dots + k\rho^{p-2}$ (where $(a, b, \dots, k)$ are integers); or (what is in one point of view more, and in another less, general)

if $\eta, \eta_1, \dots, \eta_{e-1}$ are “periods” composed with the powers of ρ (e any factor of $p-1$), then the form considered is $a\eta + b\eta_1 + \dots + h\eta_{e-1}$. For any value of p or e there is a corresponding complex theory. A number (real or complex) is in the complex theory prime or composite, according as it does not, or does, break up into factors of the form under consideration. For p a prime number under 23, if in the complex theory N is a prime, then any power of N (to fix the ideas say N^2) has no other factors than N or N^2 ; but if $p = 23$ (and similarly for higher values of p), then N may be such that, for instance, N^3 has complex factors other than N or N^3 (for $p = 23$, $N = 47$ is the first value of N , viz. 47^3 has factors other than 47 and 47^3): say N has a complex prime factor $\sqrt[3]{A}$, or we have $\sqrt[3]{A}$ as an ideal complex factor of N . Observe that by hypothesis A is not a perfect cube, viz. there is no complex number whose cube = A . In the foregoing general statement, made by way of illustration only, all reference to the complex factors of unity is purposely omitted, and the statement must be understood as being subject to correction on this account.

What precedes is by way of introduction to the account of Reuschle’s Tables (*Berliner Monatsberichte*, 1859–60), which give in the different complex theories $p = 5, 7, 11, 13, 17, 19, 23, 29$ the complex factors of the decomposable real primes up to in some cases 1000.

It should be remarked that the form of a prime factor is to a certain extent indeterminate, as the factor can without injury be modified by affecting it with a complex factor of unity; but in the tables the choice of the representative form is made according to definite rules, which are fully explained, and which need not be here referred to.

[p. 495] 53. The following synopsis is convenient:—

Theory of the p th roots	Form of real prime to mod. p	No. of factors in complex theory	Extent of table; all primes under	Equation of
$\lambda = 5$, pp. 385–401.	1 4 2, 3	4 2 (just tabulated) primes.	2500	$\eta^2 + \eta - 1 =$ $\eta^2 + \eta - 1 =$
$\lambda = 7$, pp. 494–497.	1 2, 4 3, 5	6 3 primes.	1000	$\eta^3 + \eta^2 - 2 =$ $\eta^2 + \eta + 2 =$
$\lambda = 11$, pp. 120–124.	1 3, 4, 5, 9 2 6, 7, 8, 10	10 5 2 primes.	1000	$\eta^5 + \eta^4 - 4 =$ $\eta^2 + \eta - 3 =$
$\lambda = 13$, pp. 173–179.	1	12	1000	$\eta^6 - \dots = 0$
$\lambda = 17$, pp. 724–729.	1	16	1000	$\eta^8 - \dots = 0$
$\lambda = 19$, pp. 719–724.	1	18	1000	$\eta^9 - \dots = 0$
$\lambda = 23$, pp. 737–751.	1 2, 3, 4, 6, 8, 9, 12, 13, 16, 18	22 11	1000	$\eta^{11} - \dots =$
$\lambda = 29$, pp. 761–764.	1	28	1000	$\eta^{14} - \dots =$

The foregoing synopsis of Reuschle's tables in the *Berliner Monatsberichte* was written previous to the publication of Reuschle's far more extensive work. It is

[p. 496] allowed to remain, but some explanations which were given have been struck out, and were instead given in reference to the larger work, which is

Reuschle, *Tafeln complexer Primzahlen, welche aus Wurzeln der Einheit gebildet sind*. Berlin, 4° (1875), pp. iii–vi and 1–671.

This work (the mass of calculation is perfectly wonderful) relates to the roots of unity, the degree being any prime or composite number, as presently mentioned, having all the values up to and a few exceeding 100; viz. the work is in five divisions, relating to the cases:

I. (pp. 1–171), degree any odd prime of the first 100, viz. 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97;

II. (pp. 173–192), degree the power of an odd prime 9, 25, 27, 49, 81;

III. (pp. 193–440), degree a product of two or more odd primes or their powers, viz. 15, 21, 33, 35, 39, 45, 51, 55, 57, 63, 65, 69, 75, 77, 85, 87, 91, 93, 95, 99, 105;

IV. (pp. 441–466), degree an even power of 2, viz. 4, 8, 16, 32, 64, 128;

V. (pp. 467–671), degree divisible by 4, viz. 12, 20, 24, 28, 36, 40, 44, 48, 52, 56, 60, 68, 72, 76, 80, 84, 88, the only excluded degrees being those which are the double of an odd prime, these, in fact, coming under the case where the degree is the odd prime itself.

It would be somewhat long to explain the specialities which belong to degrees of the forms II., III., IV., V.; and what follows refers only to Division I., degree an odd prime.

For instance, if $\lambda = 7$, $\lambda - 1 = 2 \cdot 3$; the factors of 6 being 6, 3, 2, 1, there are accordingly four divisions, viz. —

I. α a prime seventh root, that is, a root of $\alpha^6 + \alpha^5 + \alpha^4 + \alpha^3 + \alpha^2 + \alpha + 1 = 0$;

II. $\eta_0 = \alpha + \alpha^{-1}$, $\eta_1 = \alpha^2 + \alpha^{-2}$, $\eta_2 = \alpha^3 + \alpha^{-3}$, or η a root of

$$\eta^3 + \eta^2 - 2\eta - 1 = 0, \quad \eta_0\eta_1 = \eta_1 + \eta_2, \text{ \&c.};$$

III. $\eta_0 = \alpha + \alpha^2 + \alpha^4$, $\eta_1 = \alpha^3 + \alpha^5 + \alpha^6$, or -1 ; η a root of $\eta^2 + \eta + 2 = 0$;

IV. Real numbers.

I. $p = 7m + 1$. First, it gives for the several prime numbers of this form 29, 43, ..., 967 the congruence roots, mod. p ; for instance,

p	α	α^2	α^3	α^4	α^5	α^6
29	-5	-4	-9	-13	+7	-6
43	+11	-8	-2	+21	+16	+4.

[p. 497] This means that, if $\alpha \equiv -5 \pmod{29}$, then $\alpha^2 = 25, \equiv -4$, $\alpha^3 = 20, \equiv -9$, &c., values which satisfy the congruence $\alpha^6 + \alpha^5 + \alpha^4 + \alpha^3 + \alpha^2 + \alpha + 1 \equiv 0 \pmod{29}$.

Secondly, it gives, under the simple and the primary forms, the prime factors $f(\alpha)$ of these same numbers 29, 43, ..., 967; for instance,

p	$f(\alpha)$ simple.	$f(\alpha)$ primary.
29	$\alpha + \alpha^4 - \alpha^5$	$2 + 3\alpha - \alpha^2 + 5\alpha^3 - 7\alpha^4 + 4\alpha^5$
43	$\alpha^2 + 2\alpha^4$	$2 - 2\alpha + 2\alpha^2 - 4\alpha^3 - \alpha^4 - 5\alpha^5$.

The definition of a primary form is a form for which $f(\alpha)f(\alpha^{-1}) \equiv f(1)^2 \pmod{\lambda}$, and $f(\alpha) \equiv f(1) \pmod{-(1-\alpha)^2}$. The simple forms are also chosen so as to satisfy this last condition; thus $f(\alpha) = \alpha + \alpha^4 - \alpha^5$, then $f(1) - f(\alpha) = 1 - \alpha - \alpha^4 + \alpha^5 = (1 - \alpha)^2(1 + \alpha), \equiv 0 \pmod{(1 - \alpha)^2}$.

II. $p = 7m - 1$. First, it gives for the several prime numbers of this form 13, 41, ..., 937 the congruence roots, mod. p ; for instance,

p	η_0	η_1	η_2
13	-3	-6	-5
41	-4	+14	-11;

and secondly, it gives, under the simple and the primary forms, the prime factors $f(\eta)$ of these same numbers 13, 41, ..., 937; for instance,

p	$f(\eta)$ simple.	$f(\eta)$ primary.
13	$\eta_0 + 2\eta_1$	$3 + 3\eta_0$
41	$4 + \eta_0$	$-11 + 7\eta_0 - 7\eta_1$.

Thus $13 = (\eta_0 + 2\eta_1)(\eta_1 + 2\eta_2)(\eta_2 + 2\eta_0)$, as is easily verified; the product of the first and second factors is $4 + 3\eta_0 + 8\eta_1 + \eta_2$, and then multiplying by the third factor, the result is $42 + 29(\eta_0 + \eta_1 + \eta_2)$, = 13.

III. $p = 7m + 2$ or $7m + 4$. First, it gives for the several prime numbers of this form 2, 11, ..., 991 the congruence roots, mod. p ; for instance,

p	η_0	η_1
2	0	-1
11	4	-5;

and secondly, it gives the primary prime factors $f(\eta)$ of these same numbers; for instance,

p	$f(\eta)$
2	η_0
11	$-1 - 2\eta_1$.

[p. 498] IV. $p = 7m + 3$ or $7m + 5$. The prime numbers of these forms, viz. 3, 5, 17, 19, ..., 997, are primes in the complex theory, and are therefore simply enumerated.

The arrangement is the same for the higher prime numbers $\lambda = 23$, &c., for which ideal factors make their appearance; but it presents itself under a more complicated form. Thus $\lambda = 23$, $\lambda - 1 = 2 \cdot 11$, and the factors of 22 are 22, 11, 2, 1. There are thus four sections.

I. α a prime root, or $\alpha^{22} + \alpha^{21} + \dots + \alpha^2 + \alpha + 1 = 0$;

II. $\eta_0 = \alpha + \alpha^{-1}, \dots, \eta_{10} = \alpha^{11} + \alpha^{-11}$, or η a root of $\eta^{11} - \eta^{10} - 10\eta^9 + \dots + 10\eta^2 - 6\eta - 1 = 0$;

III. $\eta_0 = \alpha + \alpha^2 + \dots + \alpha^{11}$, $\eta_1 = \alpha^{12} + \alpha^{13} + \dots + \alpha^{22}$, or η a root of $\eta^2 + \eta + 6 = 0$;

IV. Real numbers.

I. $p = 23m + 1$. First, it gives for the prime numbers of this form 47, 139, ..., 967 congruence roots, mod. p , and also congruence roots, mod. p^3 ; these last in the form $a + bp + cp^2$, where a is given in the former table; thus first table:

p	α	α^2	α^3	...	α^{22}
47	6	-11	-19	...	+8;

and second table:

p	α	α^2	α^3	...	α^{22}
47	$-11p - 2p^2$	$+13p - 23p^2$	$+14p - 14p^2$	...	$+22p + 22p^2$.

The meaning is that, $p = 47$, the roots of the congruence

$$\alpha^{22} + \alpha^{21} + \dots + \alpha^2 + \alpha + 1 \equiv 0 \pmod{47^3}$$

are

$$\alpha = 6 + p - 2p^2, \quad \alpha^2 = -11 + 13p - 23p^2, \quad \&c.$$

Secondly, it then gives $f(\alpha)$, the actual ideal prime factor of these same primes 47, 139, ..., 967; viz. the whole of this portion of the table $\lambda = 23$, I. (2) is, having actual prime factors,

p	$f(\alpha)$
599	$\alpha + \alpha^{15} - \alpha^{17}$
691	$\alpha^2 + \alpha^{19} - \alpha^5$
829	$\alpha^{15} + \alpha^{18} + \alpha^{20}$

having ideal factors, their third powers actual,

p	$f^3(\alpha)$
47	$\alpha^2 + \alpha^4 + \alpha^6 - \alpha^{14} - \alpha^{18} + \alpha^{20}$
139	$\alpha - \alpha^2 + \alpha^3 - \alpha^6 + \alpha^{14} - 2\alpha^{18}$
277	$-\alpha^2 - \alpha^4 - \alpha^6 + \alpha^{10} + \alpha^{12} + 2\alpha^{14} - \alpha^{16} - \alpha^{18}$
461	$1 - \alpha^2 - \alpha^4 + \alpha^6 + \alpha^8 + \alpha^{12} + \alpha^{14} + \alpha^{16} + \alpha^{18} + \alpha^{20} + \alpha^{22}$
967	$-\alpha^2 - \alpha^4 + \alpha^6 - \alpha^8 - \alpha^{10} - \alpha^{12} + \alpha^{14} + \alpha^{16} \alpha^{22}$

I repeat the explanation that, for the number 47, this means $f(\alpha)f(\alpha^2)\dots f(\alpha^{22}) = 47^3$.

[p. 499] And the like further complication presents itself in the part III. of the same table, $\lambda = 23$ (not, as it happens, in part II., nor of course in the concluding part IV., which is a mere enumeration of real primes). Thus III. (1), we have congruences, $(\text{mod. } p^3)$,

$$\eta_0 \equiv p \equiv 2, \quad \eta_1 \equiv p \equiv 3, \quad \eta_0 \equiv +12, \quad \&c.;$$

and having actual prime factors,

p	$f(\eta)$
59	$-2\eta_1$
101	$1 - 4\eta_1$

and having ideal prime factors, their third powers actual,

p	$f^3(\eta)$
2	$1 - \eta_1$
3	$1 - 2\eta_1;$

⁰* Where, as presently appearing, 3 is the index of ideality or power to which the ideal factors have to be raised in order to become actual.

as regards these last the signification being

$$2^3 = (1 - \eta_0)(1 - \eta_1), \quad \eta_0 + \eta_1 = -1, \quad \eta_0\eta_1 = 6 \text{ (as is at once verified),}$$

$$3^3 = (1 - 2\eta_0)(1 - 2\eta_1);$$

but the simple numbers 2, 3 are neither of them of the form $(a + b\eta_0)(a + b\eta_1)$.

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§ 7. *Tables F. Arithmological.*

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612. Note sur une Formule d'Intégration Indéfinie

[From the *Comptes Rendus de l'Académie des Sciences de Paris*, tom. LXXVIII. (Janvier–Juin, 1874), pp. 1624–1629.]

[p. 500] En étudiant les Mémoires de M. Serret (*Journal de Liouville*, t. X., 1845) par rapport à la représentation géométrique des fonctions elliptiques, avec les remarques de M. Liouville sur ce sujet, je suis parvenu à une formule d'intégration indéfinie qui me paraît assez remarquable, savoir: en prenant θ entier positif quelconque, je dis que l'intégrale

$$\int \frac{(x+p)^{m+n-\theta-1}(x+q)^\theta dx}{x^{m+1}(x+p+q)^{n+1}}$$

a une valeur algébrique

$$(x+p)^{m+n-\theta-1}(x+p+q)^{-n}x^{-m}(A+Bx+Cx^2+\dots+Kx^{\theta-1}),$$

pourvu qu'une seule condition soit satisfaite par les quantités m, n, p, q . Cette condition s'écrit sous la forme symbolique

$$([m]p^\theta + [n]q^\theta)^\theta = 0,$$

en dénotant ainsi l'équation

$$[m]^\theta p^{\theta \cdot \theta} + \frac{\theta}{1}[m]^{\theta-1}[n]^\theta p^{\theta \cdot (\theta-1)} q^\theta + \dots + [n]^\theta q^{\theta \cdot \theta} = 0,$$

où, comme à l'ordinaire, $[m]^k$ signifie $m(m-1)\dots(m-k+1)$.

Je rappelle que les formules de M. Serret ne contiennent que des exposants entiers, et celles de M. Liouville qu'un seul exposant quelconque: la nouvelle formule contient deux exposants quelconques, m, n . Je remarque aussi l'analogie de la condition $([m]p^\theta + [n]q^\theta)^\theta = 0$ avec celle-ci

$$\frac{1}{\zeta^{m-n}} \left(\frac{d}{d\zeta} \right)^m \zeta^n (\zeta - 1)^m = 0$$

(m étant un entier positif), qui figure dans les Mémoires cités.

[p. 501] Pour démontrer la formule, j'écris

$$u = x^{-m}(A+Bx+Cx^2+\dots+Kx^{\theta-1}),$$

et aussi pour abrégé

$$X = (x+p)^{m+n-\theta+1}(x+p+q)^{-n},$$

ce qui donne

$$\frac{X}{(x+p)(x+p+q)} = (x+p)^{m+n-\theta}(x+p+q)^{-n-1}.$$

L'équation à vérifier est donc

$$Xu = \int \frac{X(x+q)^\theta dx}{x^{m+1}(x+p)(x+p+q)},$$

ou, en différentiant et divisant par X ,

$$\frac{X'}{X}u + u' = \frac{(x+q)^\theta}{x^{m+1}(x+p)(x+p+q)},$$

ou enfin

$$[(m+n-\theta+1)(x+p) - n(x+p)]u + (x+p)(x+p+q)u' = \frac{(x+q)^\theta}{x^{m+1}},$$

où u' dénote $\frac{du}{dx}$. Il ne s'agit donc que d'exprimer que cette équation ait une intégrale $u = x^{-m}(A + Bx + Cx^2 + \dots + Kx^{\theta-1})$.

En supposant que cela soit ainsi, et en effectuant la substitution, les termes en $x^{-m+\theta}$ se détruisent, et l'on obtient une équation qui contient des termes en $x^{-m-1}, x^{-m}, \dots, x^{-m+\theta-1}$, savoir $(\theta+1)$ termes. On a ainsi, entre les θ coefficients $A, B, C, \dots, K$ un système de $(\theta+1)$ équations linéaires, ce qui implique une condition entre les constantes m, n, p, q ; mais, cette condition satisfaite, les équations se réduisent à θ équations indépendantes, et les coefficients seront ainsi déterminés.

Par exemple, soit $\theta = 2$; l'équation différentielle est

$$[(m-1)p + (n-1)q + (m+n-1)x]u + [p^2 + pq + x(2p+q) + x^2]u' = x^{-m-1}(q+x)^2,$$

laquelle doit être satisfaite par $u = Ax^{-m} + Bx^{-m+1}$. Cela donne

$$\begin{vmatrix} x^{-m-1} & x^{-m} & x^{-m+1} & x^{-m+2} \\ (m-1)p + m + n - 1q)A, & (m-1)p + m + n - 1q)B, & (m-1)A & (m-1)B \\ -m(p^2 + pq)A & -(m-1)(p^2 + pq)B & -m(2p+q)A & -(m-1)(2p+q)B \\ & & -mA & \end{vmatrix} = 0.$$

[p. 502] [The text on p. 502 of the original print consists of a large multi-column determinant table whose OCR has degraded significantly; the page is largely a tabular display of the system of linear equations leading from the differential equation to the symbolic condition $([m]p^2 + [n]q^2)^2 = 0$. The page introduces the auxiliary notations $([m]p + [n]q)^1$, $([m]p + [n]q)^2$, etc., parallel to $([m]p^2 + [n]q^2)$, $([m]p^2 + [n]q^2)^2$, etc., previously explained, and shows the term-by-term computation. On a, par exemple,

$$([m]p + [n]q)^2 = [m]^2p^2 + 2[m]'[n]pq + [n]^2q^2.$$

] **[p. 503]** et ainsi de suite. Les notations $([m]p + [n]q)^2, ([m]p + [n]q)^3, \dots$ ont des significations semblables à celles de $([m]p^2 + [n]q^2)^2, ([m]p^2 + [n]q^2)^3, \dots$, auparavant expliquées. On a, par exemple,

$$([m]p + [n]q)^2 = [m]^2 p^2 + 2[m]'[n]'pq + [n]^2 q^2.$$

Considérons, par exemple, le deuxième déterminant: ceci contient trois termes en $1, 2q, q^2$ respectivement; le premier terme est

$$1 \cdot (m-1)(p^2 + pq) \cdot m(p^2 + pq),$$

c'est-à-dire

$$[m]^2 p^2 (p+q)^2;$$

le deuxième terme est

$$2q \cdot -m(p^2 + pq)[(m-1)p - nq],$$

c'est-à-dire

$$-2[m]'p(p+q)q[(m-1)p - [n]q];$$

le troisième terme est

$$q^2[(m-1)p - nq](m+1p - n-1q) - (m-1)(p^2 + pq),$$

c'est-à-dire

$$q^2[(m^2 - m)p^2 - 2mnpq + (n^2 - n)q^2] = q^2([m]p - [n]q)^2.$$

Et de même le troisième déterminant est composé de quatre termes en $1, 3q, 3q^2, q^3$ respectivement, lesquels sont les quatre termes de la première expression transformée, et ainsi pour le quatrième déterminant, etc. Au moyen de ces premières transformées, on obtient sans peine les expressions finales $([m]p^2 + [n]q^2)^2, ([m]p^2 + [n]q^2)^3, \dots$.

En écrivant $z = x + \frac{1}{2}(p+q)$ au lieu de x , et puis $\frac{1}{2}(p+q) = \alpha, \frac{1}{2}(p-q) = a$, la formule devient

$$\int \frac{(z-a)^{m+n-\theta-1}(z+a)^\theta dz}{(z-\alpha)^{m+1}(z+\alpha)^{n+1}},$$

et la valeur algébrique

$$= (z-a)^{m+n-\theta-1+1}(z+\alpha)^{-n}(z-\alpha)^{-m}(A' + B'z + \dots + K'z^{\theta-1}),$$

pourvu qu'on ait entre les quantités m, n, a, α la relation

$$\{[m](\alpha+a)^\theta + [n](\alpha-a)^\theta\}^\theta = 0.$$

En écrivant $\zeta = \frac{(\alpha + a)^\theta}{4\alpha a}$ et $\frac{(\alpha - a)^\theta}{4\alpha a} = \zeta - 1$, peut s'écrire sous la forme $\{[m]\zeta + [n](\zeta - 1)\}^\theta = 0$.

Je remarque que l'équation en ζ ne donne pas toujours pour ζ des valeurs réelles, positives et plus grandes que l'unité: par exemple, pour $\theta = 1$, on a $\zeta = \frac{-m}{n}$, valeur qui ne peut pas satisfaire à ces conditions. Je n'ai pas cherché dans quel cas ces conditions (qui ont rapport à l'application des formules à la représentation des fonctions elliptiques) subsistent.